In [ ]:
```
%pip install pdfplumber pandas numpy scikit-learn
```

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import linkage, dendrogram
```

In [2]:
```
file_path = "UCI_Credit_Card.csv"

df = pd.read_csv(file_path)

print("Dataset shape:", df.shape)
print("\nFirst five rows:")

display(df.head())
```

```
Dataset shape: (30000, 25)

First five rows:
```

	ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_0	PAY_2	PAY_3	PAY_4	...	BIL
0	1	20000.0	2	2	1	24	2	2	-1	-1	...	
1	2	120000.0	2	2	2	26	-1	2	0	0	...	
2	3	90000.0	2	2	2	34	0	0	0	0	...	
3	4	50000.0	2	2	1	37	0	0	0	0	...	
4	5	50000.0	1	2	1	57	-1	0	-1	0	...	

5 rows × 25 columns

In [3]:
```
print("Column names:")
print(df.columns.tolist())

print("\nMissing values:")
display(df.isnull().sum().sort_values(ascending=False).head())

print("\nDataset information:")
df.info()
```

```
Column names:
['ID', 'LIMIT_BAL', 'SEX', 'EDUCATION', 'MARRIAGE', 'AGE', 'PAY_0', 'PAY_2', 'PAY_
3', 'PAY_4', 'PAY_5', 'PAY_6', 'BILL_AMT1', 'BILL_AMT2', 'BILL_AMT3', 'BILL_AMT4',
'BILL_AMT5', 'BILL_AMT6', 'PAY_AMT1', 'PAY_AMT2', 'PAY_AMT3', 'PAY_AMT4', 'PAY_AMT
5', 'PAY_AMT6', 'default.payment.next.month']

Missing values:
```

```
ID           0
LIMIT_BAL    0
SEX          0
EDUCATION    0
MARRIAGE     0
dtype: int64
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 25 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   ID                          30000 non-null  int64
 1   LIMIT_BAL                   30000 non-null  float64
 2   SEX                         30000 non-null  int64
 3   EDUCATION                   30000 non-null  int64
 4   MARRIAGE                    30000 non-null  int64
 5   AGE                         30000 non-null  int64
 6   PAY_0                       30000 non-null  int64
 7   PAY_2                       30000 non-null  int64
 8   PAY_3                       30000 non-null  int64
 9   PAY_4                       30000 non-null  int64
 10  PAY_5                       30000 non-null  int64
 11  PAY_6                       30000 non-null  int64
 12  BILL_AMT1                   30000 non-null  float64
 13  BILL_AMT2                   30000 non-null  float64
 14  BILL_AMT3                   30000 non-null  float64
 15  BILL_AMT4                   30000 non-null  float64
 16  BILL_AMT5                   30000 non-null  float64
 17  BILL_AMT6                   30000 non-null  float64
 18  PAY_AMT1                    30000 non-null  float64
 19  PAY_AMT2                    30000 non-null  float64
 20  PAY_AMT3                    30000 non-null  float64
 21  PAY_AMT4                    30000 non-null  float64
 22  PAY_AMT5                    30000 non-null  float64
 23  PAY_AMT6                    30000 non-null  float64
 24  default.payment.next.month  30000 non-null  int64
dtypes: float64(13), int64(12)
memory usage: 5.7 MB
```

In [4]:

```python
excluded = {
    "ID",
    "default.payment.next.month"
}

features = ["LIMIT_BAL", "AGE", "BILL_AMT1", "PAY_AMT1"]

if len(features) < 2:
    raise ValueError(
        "The dataset does not contain enough numeric features."
    )

print("Features selected:", features)

cluster_data = df[features].copy()
```

```
cluster_data = cluster_data.replace(
    [np.inf, -np.inf], np.nan
)

cluster_data = cluster_data.fillna(
    cluster_data.median()
)

display(cluster_data.head())
```

```
Features selected: ['LIMIT_BAL', 'AGE', 'BILL_AMT1', 'PAY_AMT1']
```

	LIMIT_BAL	AGE	BILL_AMT1	PAY_AMT1
0	20000.0	24	3913.0	0.0
1	120000.0	26	2682.0	0.0
2	90000.0	34	29239.0	1518.0
3	50000.0	37	46990.0	2000.0
4	50000.0	57	8617.0	2000.0

In [5]:
```
scaler = StandardScaler()

scaled_data = scaler.fit_transform(cluster_data)

print("Standardization completed.")
print("Number of records:", scaled_data.shape[0])
```

```
Standardization completed.
Number of records: 30000
```

In [6]:
```
sample_size = min(500, len(scaled_data))

rng = np.random.default_rng(42)

sample_indices = rng.choice(
    len(scaled_data),
    size=sample_size,
    replace=False
)

sample_scaled = scaled_data[sample_indices]

linked = linkage(sample_scaled, method="ward")

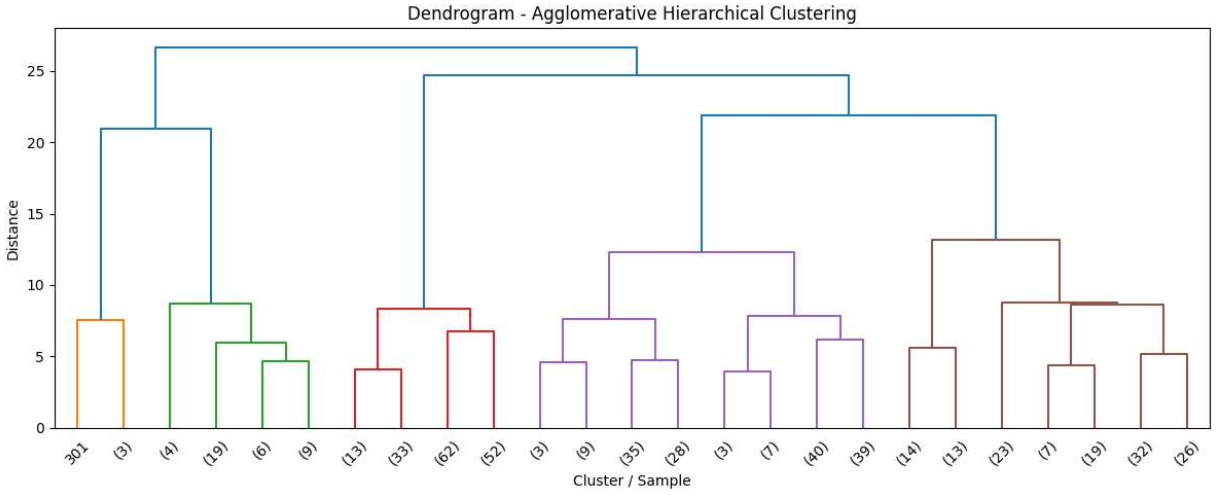
plt.figure(figsize=(12, 5))

dendrogram(
    linked,
    truncate_mode="lastp",
    p=25
)

plt.title("Dendrogram - Agglomerative Hierarchical Clustering")
```

```
plt.xlabel("Cluster / Sample")
plt.ylabel("Distance")

plt.tight_layout()
plt.show()
```



In [7]:
```
model = AgglomerativeClustering(
    n_clusters=3,
    linkage="ward"
)

labels = model.fit_predict(scaled_data)

df["Cluster"] = labels

print("Clustering completed.")
print("\nNumber of customers in each cluster:")

display(df["Cluster"].value_counts().sort_index())
```

```
Clustering completed.

Number of customers in each cluster:
Cluster
0    27049
1      245
2     2706
Name: count, dtype: int64
```

In [8]:
```
plot_df = df[features + ["Cluster"]].copy()

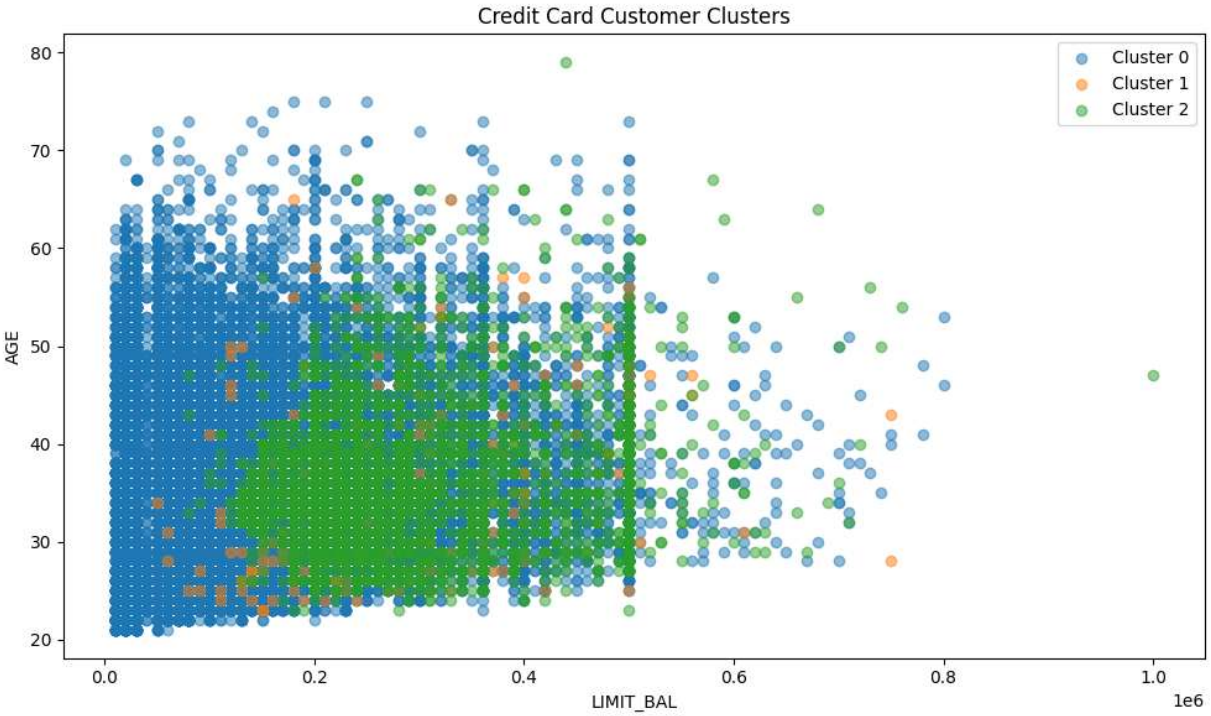
if len(plot_df) > 3000:
    plot_df = plot_df.sample(
        3000,
        random_state=42    )

plt.figure(figsize=(10, 6))

for cluster in sorted(df["Cluster"].unique()):
    subset = df[df["Cluster"] == cluster]
```

```
    plt.scatter(
        subset["LIMIT_BAL"],
        subset["AGE"],
        label=f"Cluster {cluster}",
        alpha=0.5
    )

plt.title("Credit Card Customer Clusters")
plt.xlabel("LIMIT_BAL")
plt.ylabel("AGE")
plt.legend()
plt.tight_layout()
plt.show()
```



In [9]:
```
print("Sample records with cluster labels:")

display(df[features + ["Cluster"]].head(10))

print("Average feature values for each cluster:")

display(
    df.groupby("Cluster")[features].mean().round(2)
)
```

```
Sample records with cluster labels:
```

	LIMIT_BAL	AGE	BILL_AMT1	PAY_AMT1	Cluster
0	20000.0	24	3913.0	0.0	0
1	120000.0	26	2682.0	0.0	0
2	90000.0	34	29239.0	1518.0	0
3	50000.0	37	46990.0	2000.0	0
4	50000.0	57	8617.0	2000.0	0
5	50000.0	37	64400.0	2500.0	0
6	500000.0	29	367965.0	55000.0	2
7	100000.0	23	11876.0	380.0	0
8	140000.0	28	11285.0	3329.0	0
9	20000.0	35	0.0	0.0	0

```
Average feature values for each cluster:
```

Cluster	LIMIT_BAL	AGE	BILL_AMT1	PAY_AMT1
0	153903.73	35.40	33026.02	3960.15
1	308326.53	36.18	62171.49	142564.28
2	290483.25	36.31	232131.22	10296.00